CPE 3201 - EMBEDDED SYSTEMS

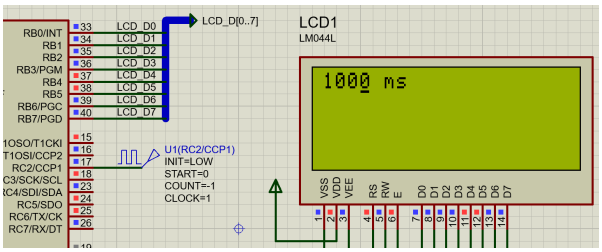
Group 4 TTh 7:30AM - 10:30AM LB 285 TC

Sarcol, Joshua S. BS CpE - 3 2026/04/06

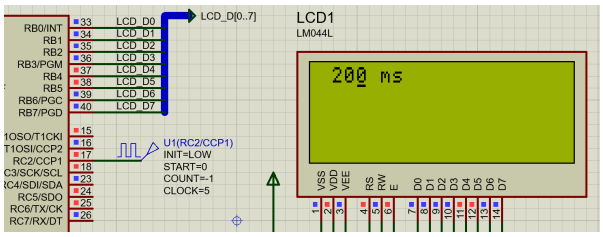
Practical Activity #4: Timers (Timer1, Timer2 and CCP Module)

PA4-6

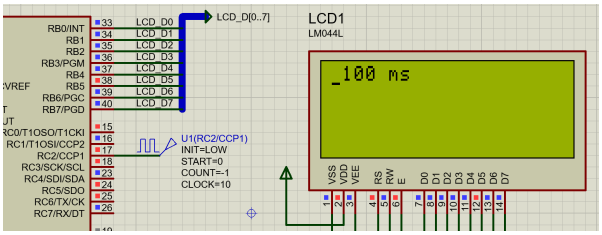
1 Hz = 1000 ms



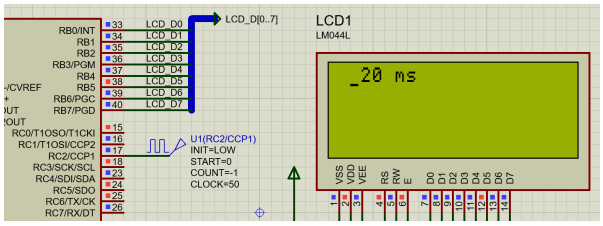
5 Hz = 200 ms



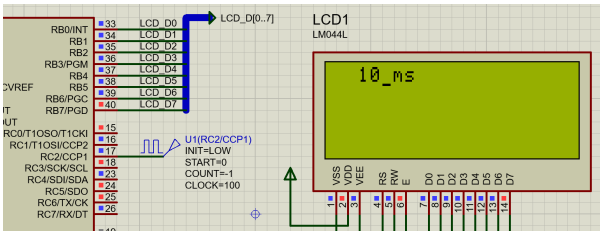
10 Hz = 100 ms



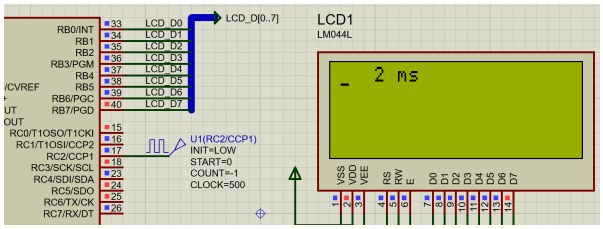
50 Hz = 20 ms



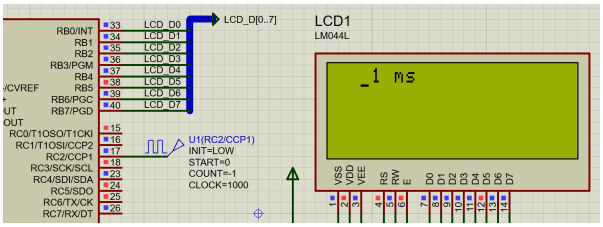
100 Hz = 10 ms



500 Hz = 2 ms



1000 Hz = 1 ms



PA 4-7

- Let "DC" be the shorthand for the 10-bit value (CCPR1L : CCP1CON<5:4>)
- assuming $F_{osc} = 4\,MHz$ ($T_{osc} = 0.25\,\mu s$) and Timer2 prescaler of 1:16

- formulas to use

$$\text{PWM period} = [(PR2) + 1] \times 4 \times T_{osc} \times \text{prescaler}_{TMR2}$$

$$\text{PWM duty cycle} = DC \times T_{osc} \times \text{prescaler}_{TMR2}$$

- 10Hz (0.1s) and 100Hz (0.01s) are not possible because the resulting PR2 values exceed 8-bit register (>255)

$0.1 = (PR2_{10} + 1) \times 4 \times 0.25\mu \times 16$

$PR2_{10} = 6249 > 255$

$0.01 = (PR2_{100} + 1) \times 4 \times 0.25\mu \times 16$

$PR2_{100} = 624 > 255$

- 500 Hz, 750 Hz and 1000 Hz will be used instead

-> At 500 Hz (2 ms)

$2m = (PR2_{500} + 1) \times 4 \times 0.25\mu \times 16$

$PR2_{500} = \boxed{124}$

=> 10% duty cycle

$2m \cdot 10\% = DC_{500/10} \times 0.25\mu \times 16$

$DC_{500/10} = \boxed{50}$

=> 50% duty cycle

$2m \cdot 50\% = DC_{500/50} \times 0.25\mu \times 16$

$DC_{500/50} = \boxed{250}$

=> 95% duty cycle

$2m \cdot 95\% = DC_{500/95} \times 0.25\mu \times 16$

$DC_{500/95} = \boxed{475}$

=> 25% duty cycle

$2m \cdot 25\% = DC_{500/25} \times 0.25\mu \times 16$

$DC_{500/25} = \boxed{125}$

=> 75% duty cycle

$2m \cdot 75\% = DC_{500/75} \times 0.25\mu \times 16$

$DC_{500/75} = \boxed{375}$

-> At 750 Hz (1.33 ms)

$1.33m = (PR2_{750} + 1) \times 4 \times 0.25\mu \times 16$

$PR2_{750} = 82.333 \approx \boxed{82}$

when set to 82

PWM period $= (82 + 1) \times 4 \times 0.25\mu \times 16$

$= 1.328\,ms \approx 753.012\,Hz$ (0.40% error)

=> 10% duty cycle

$1.328m \cdot 10\% = DC_{750/10} \times 0.25\mu \times 16$

$DC_{750/10} = 33.2 \approx \boxed{33}$

=> 50% duty cycle

$1.328\,ms \cdot 50\% = DC_{750/50} \times 0.25\mu \times 16$

$DC_{750/50} = \boxed{166}$

=> 95% duty cycle

$1.328m \cdot 95\% = DC_{750/95} \times 0.25\mu \times 16$

$DC_{750/95} = 315.4 \approx \boxed{315}$

=> 25% duty cycle

$1.328\,ms \cdot 25\% = DC_{750/25} \times 0.25\mu \times 16$

$DC_{750/25} = \boxed{83}$

=> 75% duty cycle

$1.328\,ms \cdot 75\% = DC_{750/75} \times 0.25\mu \times 16$

$DC_{750/75} = \boxed{249}$

-> At 1000 Hz (1 ms)

$1m = (PR2_{1000} + 1) \times 4 \times 0.25\mu \times 16$

$PR2_{1000} = 61.5 \approx \boxed{62}$

when set to 62

PWM period $= (62 + 1) \times 4 \times 0.25\mu \times 16$

$= 1.008\,ms \approx 992.063\,Hz$ (0.79% error)

=> 10% duty cycle

$1.008m \cdot 10\% = DC_{1000/10} \times 0.25\mu \times 16$

$DC_{1000/10} = 25.2 \approx \boxed{25}$

=> 25% duty cycle

$1.008\,ms \cdot 25\% = DC_{1000/25} \times 0.25\mu \times 16$

$DC_{1000/25} = \boxed{63}$

=> 50% duty cycle

$1.008\,ms \cdot 50\% = DC_{1000/50} \times 0.25\mu \times 16$

$DC_{1000/50} = \boxed{126}$

=> 75% duty cycle

$1.008\,ms \cdot 75\% = DC_{1000/75} \times 0.25\mu \times 16$

$DC_{1000/75} = \boxed{189}$

=> 95% duty cycle

$1.008m \cdot 95\% = DC_{1000/95} \times 0.25\mu \times 16$

$DC_{1000/95} = 239.4 \approx \boxed{239}$

- Result summary

-> PR2 values

	500 Hz	750 Hz	1000 Hz
PR2	124 (0x7C)	82 (0x52)	62 (0x3E)

-> CCPR1L : CCP1CON <5:4> values

	10%	25%	50%	75%	95%
500 Hz	50 ($00\ 0011\ 0010_2$)	125 ($00\ 0111\ 1101_2$)	250 ($00\ 1111\ 1010_2$)	375 ($01\ 0111\ 0111_2$)	475 ($01\ 1101\ 1011_2$)
750 Hz	33 ($00\ 0010\ 0001_2$)	83 ($00\ 0101\ 0011_2$)	166 ($00\ 1010\ 0110_2$)	249 ($00\ 1111\ 1001_2$)	315 ($01\ 0011\ 1011_2$)
1000 Hz	25 ($00\ 0001\ 1001_2$)	63 ($00\ 0011\ 1111_2$)	126 ($00\ 0111\ 1110_2$)	186 ($00\ 1011\ 1010_2$)	239 ($00\ 1110\ 1111_2$)